

Developer Guide (Part 2): Financial Markets & Market Data Fundamentals

Version: 1.0

Audience: Software Engineers, Quant Developers, Data Engineers

Prerequisites: Developer Guide (Part 1): Introduction & System Overview

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1. Purpose

A quantitative strategy is only as good as the data it consumes.

Before implementing trading algorithms, developers must understand:

- How financial markets operate.
- Who participates in the markets.
- What types of market data are available.
- How that data is generated and distributed.
- Common data quality issues.

This document introduces the market concepts required to build robust quantitative trading systems.

2. Financial Market Overview

A financial market is a marketplace where buyers and sellers exchange financial instruments.

Markets perform several essential functions:

- Price discovery
- Liquidity provision
- Capital allocation
- Risk transfer

Electronic exchanges continuously match buy and sell orders based on predefined matching rules.

3. Asset Classes

Different quantitative strategies operate on different asset classes.

Asset Class	Description	Examples
Equities	Ownership in companies	Apple, Microsoft
ETFs	Basket of securities	SPY, QQQ
Fixed Income	Debt instruments	Government Bonds
Commodities	Physical assets	Gold, Oil
Forex	Currency pairs	EUR/USD
Futures	Contracts to buy/sell later	Crude Oil Futures
Options	Rights without obligation	Call, Put
Cryptocurrency	Digital assets	Bitcoin

4. Market Participants

Understanding who trades is important because each participant has different objectives.

Retail Investors

- Individual traders
 - Smaller order sizes
 - Long-term investing or speculation
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Institutional Investors

- Mutual funds
- Pension funds
- Insurance companies

Typically trade millions of dollars daily.

Hedge Funds

Focus on alpha generation using quantitative models.

Proprietary Trading Firms

Trade using the firm's capital.

Examples include firms specializing in high-frequency and algorithmic trading.

Market Makers

Continuously provide liquidity by quoting bid and ask prices.

5. Order Types

A trading system sends orders to an exchange through a broker.

Market Order

Executes immediately at the best available price.

Advantages

- Guaranteed execution

Disadvantages

- Price uncertainty
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Limit Order

Executes only at a specified price or better.

Advantages

- Price control

Disadvantages

- May not execute
-

Stop Order

Becomes active after reaching a trigger price.

Used primarily for risk management.

Stop Limit Order

Combines stop and limit conditions.

Provides price protection while controlling execution.

6. Exchange vs OTC Markets

Exchange-Traded Markets

Examples

- NYSE
- NASDAQ
- NSE
- BSE

Characteristics

- Transparent pricing
 - Centralized matching
 - Regulatory oversight
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Over-the-Counter (OTC)

Trades occur directly between counterparties.

Examples

- Foreign Exchange
 - Some derivatives
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7. Market Sessions

Markets generally have:

- Pre-market
- Regular session
- After-hours session

Liquidity and volatility vary significantly across sessions.

8. What is Market Data?

Market data represents the state of the financial market at a given point in time.

It includes:

- Prices
 - Volumes
 - Orders
 - Trades
 - Corporate actions
 - Fundamental information
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9. Types of Market Data

Historical Data

Used for:

- Research
 - Backtesting
 - Machine learning
-

Real-Time Data

Used for:

- Live trading
 - Monitoring
 - Signal generation
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Reference Data

Examples:

- Symbol mappings
 - Company names
 - Exchange codes
-

Fundamental Data

Includes:

- Revenue
 - Earnings
 - Balance sheets
 - Financial ratios
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10. OHLCV Data

OHLCV is the most common data format in quantitative research.

Field	Description
Open	First traded price
High	Highest price
Low	Lowest price
Close	Final traded price
Volume	Number of shares traded

Example

Time	Open	High	Low	Close	Volume
09:15	150	152	149	151	250000

Most beginner strategies are built using OHLCV data.

11. Tick Data

Tick data records every market event.

Each record may contain:

- Timestamp

- Trade price
- Quantity
- Trade direction

Tick data enables:

- High-frequency trading
 - Order flow analysis
 - Microstructure research
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12. Order Book

The order book shows all active buy and sell orders.

Example

SELL

151.20 400

151.10 250

151.00

150.95 300

150.90 500

BUY

The difference between best bid and best ask is called the **bid-ask spread**.

Strategies such as market making rely heavily on order book data.

13. Corporate Actions

Corporate actions affect historical prices and must be accounted for.

Examples:

- Stock splits
- Dividends
- Bonus issues
- Rights issues
- Mergers
- Acquisitions

Failing to adjust for corporate actions can produce incorrect backtesting results.

14. Data Quality Challenges

Reliable strategies require clean data.

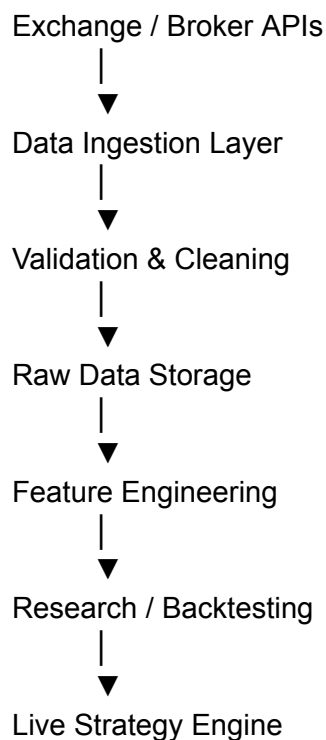
Common issues include:

- Missing values
- Duplicate records
- Incorrect timestamps
- Outliers
- Survivorship bias
- Look-ahead bias
- Time zone inconsistencies

Developers should implement validation pipelines before using any dataset.

15. Market Data Architecture

A simplified architecture for handling market data is shown below:



Each stage should be modular, monitored, and scalable to support both historical research and live trading.

Key Takeaways

- Market data is the foundation of every quantitative strategy.
 - Developers should understand the characteristics and limitations of each data type.
 - Data quality directly impacts strategy performance.
 - Robust ingestion, validation, and storage pipelines are essential for production-grade systems.
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References

Books

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Research

- Qlib: An AI-oriented Quantitative Investment Platform (Microsoft Research)
- FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading

Open Source

- Microsoft Qlib
- QuantConnect Lean
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